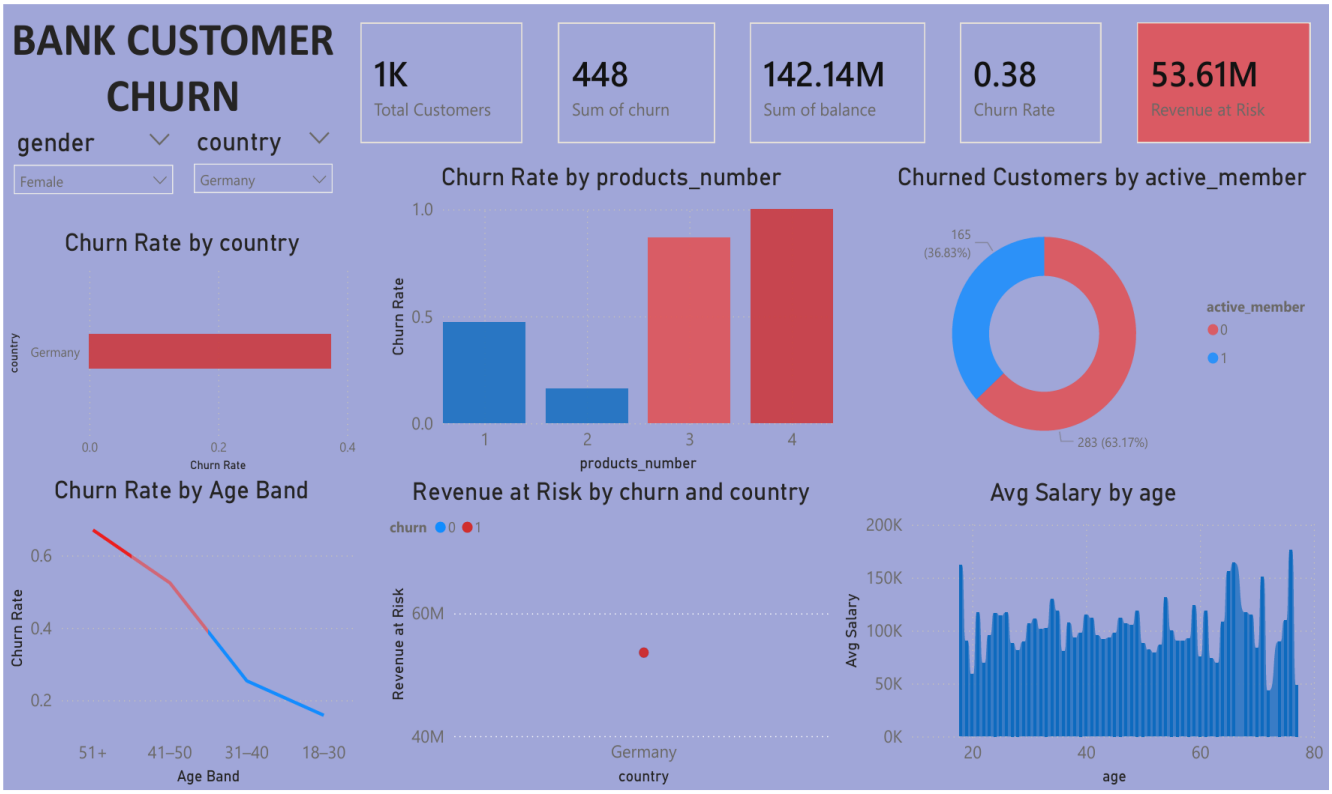
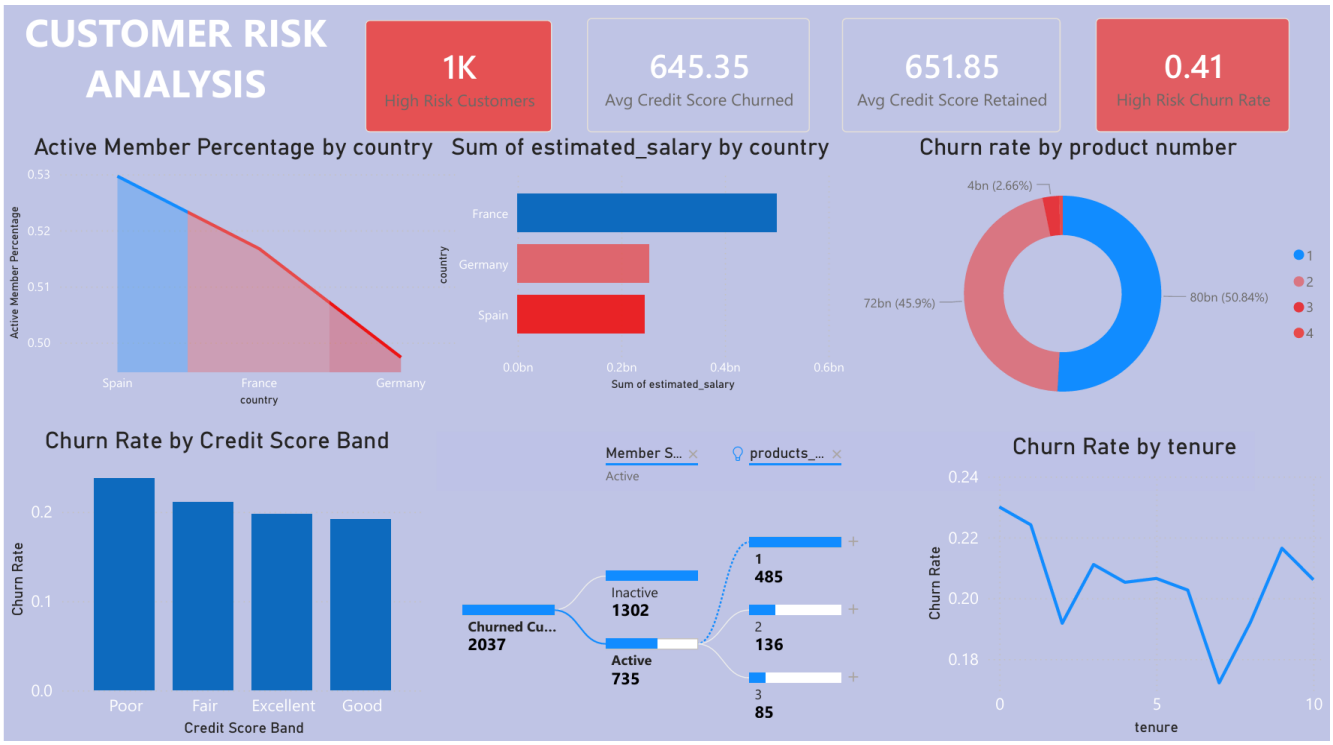
BANK CUSTOMER CHURN ANALYSIS

Submitted by: - Khushi Bandpatte

Technologies Used

- Python
- MySQL
- Power BI
- DAX




CUSTOMER RISK ANALYSIS
1K
High Risk Customers
645.35
Avg Credit Score Churned
651.85
Avg Credit Score Retained
0.41
High Risk Churn Rate
Active Member Percentage by country
0.53
0.52
0.51
0.50
Active Member Percentage
Spain
France
Germany
country
Sum of estimated_salary by country
France
Germany
Spain
country
0.0bn
0.2bn
0.4bn
0.6bn
Sum of estimated_salary
Churn rate by product number
4bn (2.66%)
72bn (45.9%)
80bn (50.84%)
1
2
3
4
Churn Rate by Credit Score Band
0.2
0.1
0.0
Churn Rate
Poor
Fair
Excellent
Good
Credit Score Band
Member S...
Active
products_...
Churned Cu...
2037
Inactive
1302
Active
735
1
485
2
136
3
85
Churn Rate by tenure
0.24
0.22
0.20
0.18
Churn Rate
0
5
10
tenure

1. Introduction:-

Customer churn refers to the percentage of customers who discontinue their relationship with a business over a given period. In the banking sector, customer retention is essential because acquiring new customers is significantly more expensive than retaining existing ones.

This project aims to identify the factors contributing to customer attrition and provide actionable recommendations that can help banks reduce customer loss and improve profitability.

2. Problem Statement: -

Banks maintain large volumes of customer information, including demographics, financial activity, product usage, and transaction history. Despite this information, many institutions struggle to identify the factors responsible for customer attrition.

The objective of this project is to analyze customer data, identify high-risk customer groups, and develop an interactive dashboard that supports informed decision-making.

3. Objectives: -

The major objectives of this project are:

- Calculate the customer churn rate.
- Identify the factors influencing customer churn.
- Analyze customer behavior patterns.
- Determine high-risk customer groups.
- Estimate the revenue associated with customer attrition.
- Provide business recommendations.

4. Dataset Description

Attribute	Description
customer_id	Unique customer identifier
credit_score	Customer credit score
country	Customer location
gender	Customer gender
age	Customer age
tenure	Years of association with the bank
balance	Account balance
products_number	Number of products used
credit_card	Credit card ownership status
active_member	Customer activity status
estimated_salary	Estimated annual salary
churn	Customer retention status

5. Data Collection and Preparation

The dataset was imported into MySQL and connected to Power BI for analysis.The following preprocessing steps were performed:

- Data cleaning
- Data transformation
- Column renaming
- Data type conversion
- Handling null values
- Creating age bands
- Creating credit score bands
- Creating high-risk customer segments

6. Database Architecture:-

The data was stored inside the **churn_prediction** database.

The following tables were created:

- bank_customer_churn
- customer_demographics
- account_details

The primary key used in the dataset was **customer_id**.

7. DAX Measures: -

Total Customers

Total Customers =

COUNTROWS(bank_customer_churn)

Churned Customers

```
Churned Customers =
CALCULATE(
   COUNTROWS(bank_customer_churn),
   bank_customer_churn[churn] = 1
)
```

Churn Rate

```
Churn Rate =
DIVIDE(
   [Churned Customers],
   [Total Customers]
)
```

Revenue at Risk

Revenue at Risk =

```
CALCULATE(
    SUM(bank_customer_churn[balance]),
    bank_customer_churn[churn] = 1
)
```

High-Risk Customers

```
High Risk Customers =
CALCULATE(
    COUNTROWS(bank_customer_churn),
    bank_customer_churn[High Risk Segment] = "High Risk"
)
```

8. Dashboard Overview: -

The dashboard consists of two pages.

Page 1 – Executive Dashboard

This page provides a high-level summary of customer behavior.

Visuals included

- KPI cards
- Country-wise churn analysis
- Product-wise churn analysis
- Revenue-at-risk analysis
- Age-band analysis
- Customer activity analysis

Page 2 – Customer Risk Analysis

This page focuses on identifying the reasons behind customer attrition.

Visuals included

- High-risk customer analysis
- Credit score analysis

- Decomposition tree
- Country-wise activity analysis
- Tenure analysis
- Product analysis

9. Key Findings: -

- **Country analysis: -** Germany exhibited the highest churn rate among all countries.
- **Product analysis: -** Customers using three or more products showed significantly higher churn rates.
- **Customer activity analysis: -** Inactive customers were considerably more likely to discontinue their relationship with the bank.
- **Age analysis: -** Customers between 41 and 60 years of age showed the highest probability of churn.
- **Revenue analysis: -** A significant amount of revenue was found to be associated with customers who were at risk of leaving.

10. Business Recommendations:-

- **Recommendation 1:-** Develop customer retention strategies for inactive customers.
- **Recommendation 2: -** Provide personalized offers to high-risk customers.
- **Recommendation 3:-** Improve customer engagement and loyalty programs.
- **Recommendation 4: -** Strengthen customer support services.
- **Recommendation 5: -**Implement predictive analytics to identify high-risk customers in advance.

11. Conclusion: -

This project successfully analyzed customer churn behavior using data visualization techniques and business intelligence tools. The results revealed that customer activity status, age, country, and product ownership play an essential role in customer retention.

The dashboard enables stakeholders to identify high-risk customers, monitor business performance, and make informed decisions based on data.